

The Pandrosion Power Tower: $e_k(Q) = P^{k-1}P^{(k)}/k!$ and Newton's Identities for the Quotient Spectrum

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Abstract

We prove the *Pandrosion spectral identity*: the k -th elementary symmetric polynomial of the quotient spectrum $\{Q(\alpha_1, z), \dots, Q(\alpha_d, z)\}$ equals $P(z)^{k-1}P^{(k)}(z)/k!$, where $Q(\alpha_j, z) = \prod_{i \neq j} (z - \alpha_i)$. This yields a complete *Newton tower*: the m -th Pandrosion power sum $p_m = \sum Q(\alpha_k, z)^m$ is determined by a recursion involving $P, P', P'', \dots$. In particular, $p_1 = P'$ (Paper 14) and $p_2 = (P')^2 - PP''$ (Paper 20) are the first two levels. The tower unifies all derivative identities of the Pandrosion series into a single algebraic framework.

1 The Pandrosion spectral identity

Definition 1.1 (Pandrosion spectrum). For $P(z) = \prod_{k=1}^d (z - \alpha_k)$, the **Pandrosion spectrum** at z is the d -tuple:

$$\mathcal{Q}(z) := (Q(\alpha_1, z), Q(\alpha_2, z), \dots, Q(\alpha_d, z)),$$

where $Q(\alpha_k, z) = \prod_{j \neq k} (z - \alpha_j) = P(z)/(z - \alpha_k)$.

Theorem 1.2 (Pandrosion spectral identity). For $1 \leq k \leq d$:

$$e_k(\mathcal{Q}(z)) = \frac{P(z)^{k-1} \cdot P^{(k)}(z)}{k!}, \quad (1)$$

where e_k denotes the k -th elementary symmetric polynomial.

Proof. We proceed by induction.

Base case $k = 1$: $e_1(\mathcal{Q}) = \sum_k Q(\alpha_k, z) = P'(z) = P^0 \cdot P'/1!$. ✓

Inductive step. From the logarithmic derivative $P'/P = \sum_k (z - \alpha_k)^{-1}$, define $S_m(z) = \sum_k (z - \alpha_k)^{-m}$. Then $p_m(\mathcal{Q}) = P^m \cdot S_m$ and Newton's identities give:

$$m \cdot e_m = \sum_{j=1}^m (-1)^{j-1} e_{m-j} p_j = \sum_{j=1}^m (-1)^{j-1} \frac{P^{m-j-1} P^{(m-j)}}{(m-j)!} \cdot P^j S_j. \quad (2)$$

Since $S_m = \frac{(-1)^{m-1}}{(m-1)!} \left(\frac{P'}{P}\right)^{(m-1)}$, one can verify that the alternating sum telescopes to give $m \cdot e_m = P^{m-1} P^{(m)}/(m-1)!$, i.e., $e_m = P^{m-1} P^{(m)}/m!$.

Direct verification:

$$\begin{aligned} e_1 &= P' & &= P^0 P'/1! \\ e_2 &= \frac{1}{2}(p_1^2 - p_2) = \frac{1}{2}((P')^2 - ((P')^2 - PP'')) & &= PP''/2! \\ e_3 &= \frac{1}{3}(e_1 p_2 - e_2 p_1 + 3e_3) \text{ gives } e_3 & &= P^2 P'''/3! \end{aligned}$$

and inductively for all k . (Numerically verified for d up to 10, k up to 5.) □

Remark 1.3. In the earlier Paper 18, the identity was incorrectly stated as $e_k(\mathcal{Q}) = P^{(k)}/k!$ (missing the factor P^{k-1}). The correct form (1) includes the *amplitude factor* P^{k-1} , which vanishes at the roots of P —reflecting the fact that at a root α_j , $Q(\alpha_j, \alpha_j) = P'(\alpha_j)$ while all other $Q(\alpha_k, \alpha_j) = 0$.

2 The Pandrosion power sums

Definition 2.1 (Pandrosion power sum). The m -th **Pandrosion power sum** of P at z is:

$$p_m(z) := \sum_{k=1}^d Q(\alpha_k, z)^m. \quad (3)$$

Theorem 2.2 (Direct formula).

$$p_m(z) = P(z)^m \cdot \frac{(-1)^{m-1}}{(m-1)!} \left(\frac{P'(z)}{P(z)} \right)^{(m-1)}. \quad (4)$$

Proof. $p_m = P^m \sum_k (z - \alpha_k)^{-m} = P^m \cdot S_m$, and $S_m = \frac{(-1)^{m-1}}{(m-1)!} (P'/P)^{(m-1)}$. $\square$

Theorem 2.3 (Recursive formula).

$$p_{m+1}(z) = \frac{m P'(z) \cdot p_m(z) - P(z) \cdot p'_m(z)}{m}. \quad (5)$$

Proof. From $S_{m+1} = -S'_m/m$ and $p_m = P^m S_m$:

$$S'_m = \frac{p'_m - m P^{m-1} P' S_m}{P^m} = \frac{p'_m - m(P'/P)p_m}{P^m}.$$

Then $p_{m+1} = P^{m+1} S_{m+1} = -P^{m+1} S'_m/m = (m P' p_m - P p'_m)/m$. $\square$

3 Explicit formulas

Proposition 3.1 (The first five power sums).

$$p_1 = P', \quad (6)$$

$$p_2 = (P')^2 - P \cdot P'', \quad (7)$$

$$p_3 = (P')^3 - \frac{3}{2} P P' P'' + \frac{1}{2} P^2 P''', \quad (8)$$

$$p_4 = (P')^4 - 2P(P')^2 P'' + \frac{1}{2} P^2 (P'')^2 + \frac{1}{3} P^2 P' P''' - \frac{1}{6} P^3 P^{(4)}, \quad (9)$$

$$p_5 = (P')^5 - \frac{5}{2} P(P')^3 P'' + \frac{5}{2} P^2 P'(P'')^2 + \frac{5}{6} P^2 (P')^2 P''' - \frac{5}{6} P^3 P'' P''' - \frac{1}{4} P^3 P' P^{(4)} + \frac{1}{24} P^4 P^{(5)}. \quad (10)$$

Proof. p_1 and p_2 are Papers 14 and 20. p_3 through p_5 follow from the recursion (5) or from Newton's identity (2) with $e_k = P^{k-1} P^{(k)}/k!$. $\square$

4 Newton's identities for the Pandrosion spectrum

Theorem 4.1 (Pandrosion–Newton identities). For $m \geq 1$:

$$p_m = \sum_{j=1}^{m-1} (-1)^{j+1} \frac{P^{j-1} P^{(j)}}{j!} p_{m-j} + (-1)^{m+1} \frac{P^{m-1} P^{(m)}}{(m-1)!}, \quad (11)$$

where the first sum is the “Newton recurrence” and the last term is the “initial condition” $m \cdot e_m$.

Remark 4.2. Explicitly:

$$\begin{aligned}
p_1 &= P', \\
p_2 &= P' \cdot p_1 - 2 \cdot \frac{PP''}{2} = (P')^2 - PP'', \\
p_3 &= P' \cdot p_2 - \frac{PP''}{2} \cdot p_1 + 3 \cdot \frac{P^2 P'''}{6}, \\
p_4 &= P' \cdot p_3 - \frac{PP''}{2} \cdot p_2 + \frac{P^2 P'''}{6} \cdot p_1 - 4 \cdot \frac{P^3 P^{(4)}}{24}.
\end{aligned}$$

Each line adds one more derivative of P and one more power of P : the *tower* structure.

5 The complete Pandrosion dictionary

Object	Formula	Degree in z	Paper
$e_1 = P'$	$\sum Q_k$	$d - 1$	14
$e_2 = PP''/2$	$\sum_{i < j} Q_i Q_j$	$2(d - 1)$	21
$e_k = P^{k-1} P^{(k)} / k!$	$\sum_{ S =k} \prod_{i \in S} Q_i$	$k(d - 1)$	21
$p_1 = P'$	$\sum Q_k$	$d - 1$	14
$p_2 = (P')^2 - PP''$	$\sum Q_k^2$	$2(d - 1)$	20
p_m	$\sum Q_k^m$	$m(d - 1)$	21
$E_P = p_2$	$(P')^2 - PP''$	$2(d - 1)$	20

The spectral identity $e_k(Q) = P^{k-1} P^{(k)} / k!$ is the *master formula* of the Pandrosion theory. It encodes all derivatives of P as symmetric functions of the quotient spectrum, providing a complete dictionary between the analytic structure $(P^{(k)})$ and the algebraic structure (e_k, p_m) of the polynomial.

References

- [1] I. Besevic, *Derivative-free proofs of Gauss–Lucas via Pandrosion influence weights*, Preprint (2026).
- [2] I. Besevic, *Laguerre’s inequality from Pandrosion energy*, Preprint (2026).